EIEN 133 HW 3

Dylan Trinity

1. $y(n) = a\,y(n-1) + x(n)$

a.

$x(n) = A\delta(n)$ $y(-1)=0$

n	$x(n)$	$y(n-1)$	$y(n)$
0	A	0	A
1	0	aA	aA
2	0	a^2A	a^2A
3	0	a^3A	a^3A

$$y(n) = Aa^n u(n)$$

$x(n) = Bu(n)$, $y(-1)=0$

n	$x(n)$	$y(n-1)$	$y(n)$
0	B	0	B
1	B	aB	$aB+B$
2	B	$a(aB+B)$	$a^2B+Ba+B$
3	B	$a(a^2B+Ba+B)$	$B(a^3+a^2+a+1)$

B/c

$$\sum_{k=0}^{M} a^k = \begin{cases} \dfrac{1-a^{M+1}}{1-a} & a \neq 1 \\ M+1 & a=1 \end{cases}$$

$$y(n) = B\sum_{k=0}^{n} a^k = Bu(n)\left(\frac{1-a^{n+1}}{1-a}\right) \quad \text{for } a \neq 1$$

$$y(n) = Bu(n)(n+1) \quad \text{for } a=1$$

2.

Monthly = \$832258.64

vs

Annually = \$797266.17

Monthly gives ~35k more by the end of the period, but annually gives more returns at the start.

2.

a.

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Final amount y[359] = 832258.64
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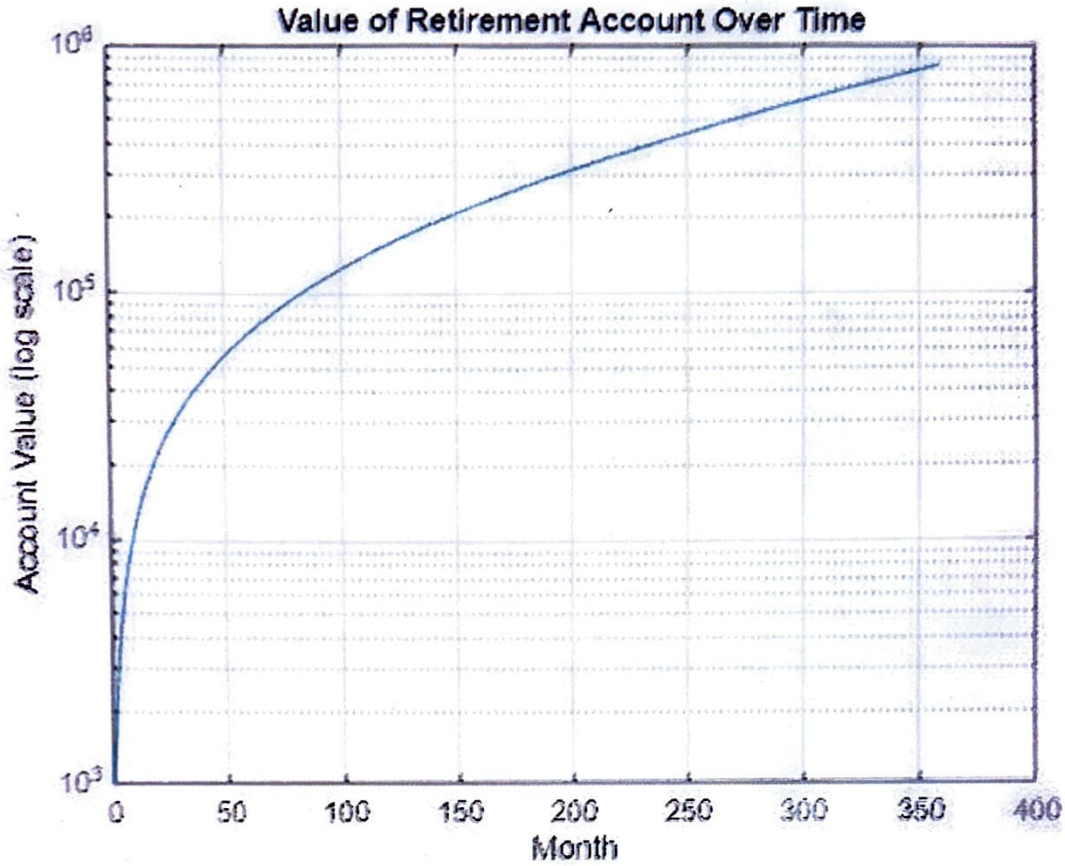

b.

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Final amount y[29] = 797266.17
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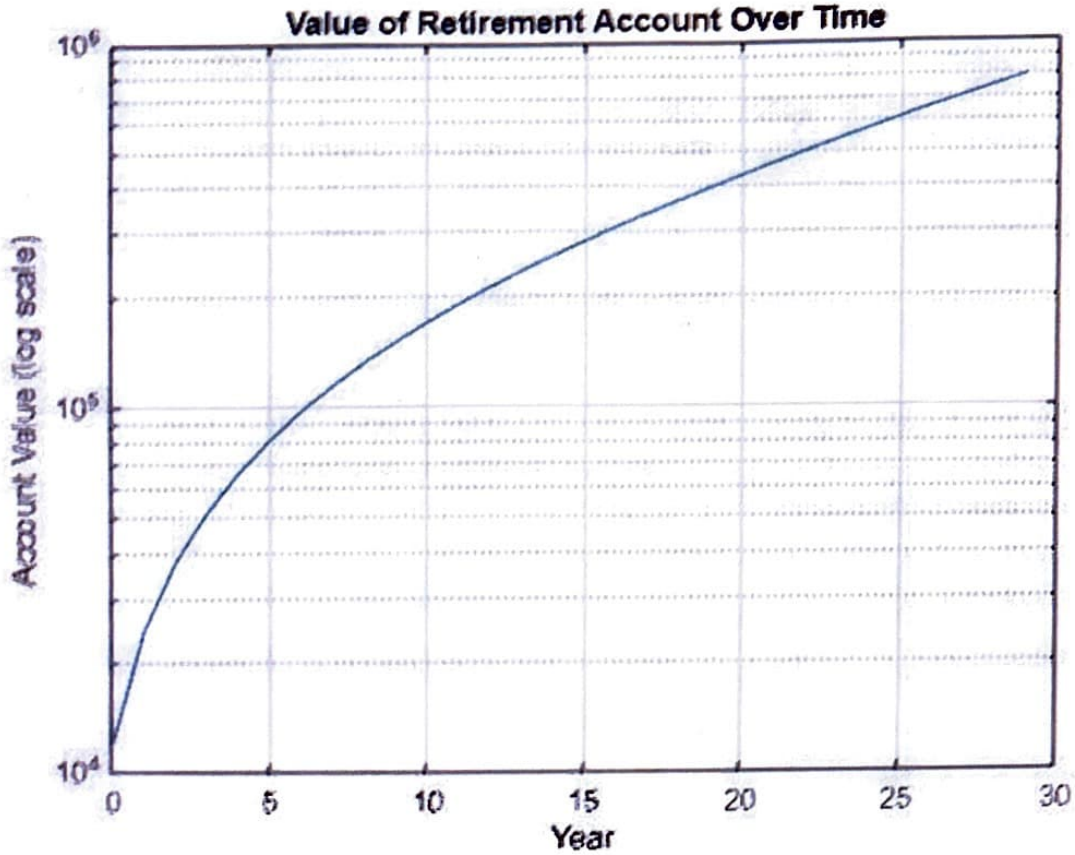

Value of Retirement Account Over Time
Account Value (log scale)
10^6
10^5
10^4
0
5
10
15
20
25
30
Year

3.

a. $h_1(n) = \delta(n) - 5\delta(n-1) + \delta(n-2)$

$1 - 5e^{-jw} + e^{-2jw}$

$e^{-jw}(e^{jw} - 5 + e^{-jw}) = \boxed{e^{-jw}(2\cos w - 5)}$

mag $= |2\cos w - 5|$

delay $= 1$ (from e^{-jw})

b. $h_2(n) = \delta(n+1) - 5\delta(n) + \delta(n-1)$

$h_2(n) = e^{jw} - 5 + e^{-jw}$

$h_2(n) = 2\cos w - 5$ mag $= |2\cos w - 5|$

delay $= 0$

c. $h_3(n) = \delta(n) - 5\delta(n-2) + \delta(n-4)$

$1 - 5e^{-2jw} + e^{-4jw}$

$e^{-2jw}(e^{2jw} - 5 + e^{-2jw})$

$h_3(n) = e^{-2jw}(2\cos(2w) - 5)$

mag $= |2\cos(2w) - 5|$

delay $= 2$

a and b have the same mag but diff delays

4.

$$\delta(n-k) \xleftrightarrow{DTFT} e^{-jwk}$$

$$X(e^{jw}) = 8\cos(w) - 9$$

$$\downarrow$$

$$4e^{-jw} + 4e^{jw} - 9$$

$$\boxed{X(e^{jw}) = 4\delta(n-1) + 4\delta(n+1) - 9\delta(n)}$$

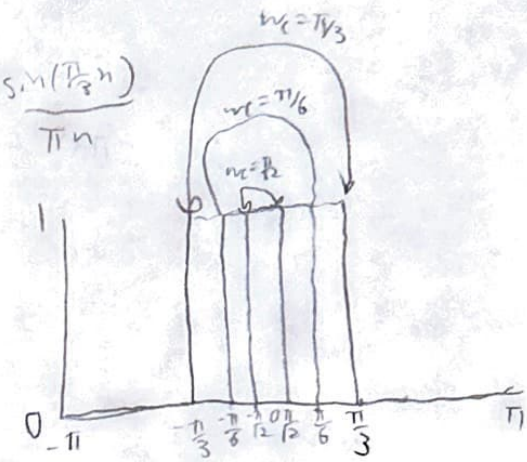
5. $H_{LP}(n) = \frac{w_c}{\pi} \frac{\sin(w_c n)}{w_c n}$

a. $w_c = \frac{\pi}{3}$, $h_{LP}(n) = \frac{1}{3} \cdot \frac{\sin(\frac{\pi}{3}n)}{\pi n/3} = \frac{\sin(\frac{\pi}{3}n)}{\pi n}$

$w_c = \frac{\pi}{6}$, $\frac{\sin(\frac{\pi}{6}n)}{\pi n}$

$w_c = \frac{\pi}{12}$, $\frac{\sin(\frac{\pi}{12}n)}{\pi n}$

Idea



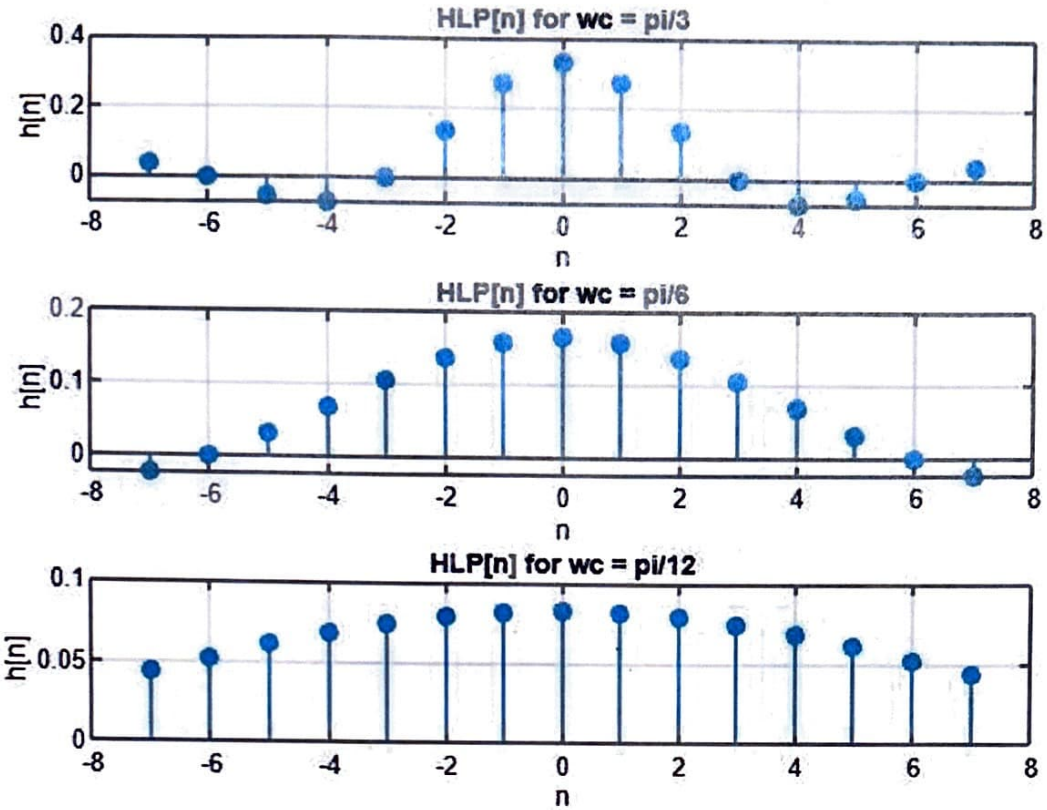
b. $n=0$ gives undefined, but taking limits shows we approach $\frac{1}{3}, \frac{1}{6}, \frac{1}{12}$

$\pi/3$, n is a multiple of 3 gives $H_{LP}=0$

$\pi/6$, n is a multiple of 6 gives $H_{LP}=0$

$\pi/12$, n is a multiple of 12 gives $H_{LP}=0$

c.


HLP[n] for wc = pi/3
h[n]
n
HLP[n] for wc = pi/6
h[n]
n
HLP[n] for wc = pi/12
h[n]
n

6.

a. $x(n) = a^n u(n)$

$$X(w) = \sum_{n=-\infty}^{\infty} x(n) e^{-jwn}$$

$$X(w) = \sum_{n=0}^{\infty} a^n e^{-jwn}$$

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

$$X(w) = \sum_{n=0}^{\infty} (a e^{-jw})^n$$

$$\boxed{X(w) = \frac{1}{1 - a e^{-jw}}}$$

b. $x(n) = a^n u(n-2)$

$$X(w) = \sum_{n=2}^{\infty} a^n e^{-jwn} = \sum_{n=0}^{\infty} a^n e^{-jwn} - (1 + a e^{-jw})$$

$$\frac{1}{1 - a e^{-jw}} - 1 - a e^{-jw}$$

$$\frac{1}{1 - a e^{-jw}} - \frac{(1 - a e^{-jw})}{1 - a e^{-jw}} - \frac{a e^{-jw}(1 - a e^{-jw})}{(1 - a e^{-jw})}$$

$$\frac{a e^{-jw} - a e^{-jw} + a^2 e^{-j2w}}{1 - a e^{-jw}}$$

$$\boxed{X(w) = \frac{a^2 e^{-j2w}}{1 - a e^{-jw}}}$$

$x(n) = a^{n-2} u(n-2)$

just $a^{-2}\left(a^{n} u(n-2)\right)$ which we know is $\frac{a^{2} e^{-j2w}}{1-ae^{-jw}}$

$$\boxed{X(w) = \frac{e^{-j2w}}{1-ae^{-jw}}}$$

7. $x(n) = a^{n} u(-n-1)$

$$X(w) = \sum_{-\infty}^{\infty} x(n) e^{-jwn}$$

$$\sum_{n=-\infty}^{-1} a^{n} e^{-jwn} \rightarrow \sum_{m=1}^{\infty} a^{-m} e^{jwm} = \sum_{m=1}^{\infty} \left(a^{-1} e^{jw}\right)^{m} \qquad m=0 \Rightarrow 1$$

$$\sum_{m=0}^{\infty} \left(a^{-1} e^{jw}\right)^{m} - 1$$

$$\frac{1}{1-a^{-1}e^{jw}} - 1 = X(w)$$

$$X(w) = \frac{1-(1-a^{-1}e^{jw})}{1-a^{-1}e^{jw}} = \boxed{\frac{a^{-1}e^{jw}}{1-a^{-1}e^{jw}}}$$

Exists when $|a|>1$ vs 6a where $|a|<1$ to exist

6a is causal

7 is anticausal

(looked this up online, may be wrong)